

Attendify Codebase Guide

Updated for the current Expo + Supabase project structure and workflow.

Project overview

Webcapz is a role-based attendance management app built with Expo Router, Supabase, and React Query. The current implementation supports three roles: Admin, Student, and Lecturer, with role-gated navigation and attendance recording flows.

Current app metadata

Name	Webcapz
Package	expo-router/entry
Version	1.0.0
Platform	iOS + Android + web
Orientation	Portrait
Architecture	React Native with Expo SDK 54 and New Architecture enabled

Core stack

- Expo SDK 54, Expo Router, React Native 0.81.5, TypeScript
- Supabase Auth + Postgres + Row Level Security
- TanStack React Query for client state and cached data
- expo-camera for QR scanning and react-native-qrcode-svg for QR generation
- expo-image-picker, expo-file-system, and Supabase Storage for profile photos

Current project structure

- app/ — Expo Router screens for login, onboarding, admin routes, and member routes
- app/(admin)/ — admin dashboard, scan view, attendance views, students, lecturers, and account creation
- app/(member)/ — member home, scan, history, profile, and my-scans screens
- components/ — shared UI components, QR card, scanner, avatar picker, and list items
- hooks/ — auth and data hooks for profiles, attendance, onboarding, and admin flows
- services/ — Supabase request wrappers for auth, profiles, attendance, admin actions, and storage
- lib/ — Supabase client, config, date helpers, and query client setup
- supabase/ — SQL schema and create-user Edge Function
- types/ — shared database and role typing

Role-based navigation

- Admins see Dashboard, Scan, Attendance, Students, and Lecturers tabs.

- Members use Home, History, Profile, and, for lecturers, Scan and My Scans tabs.
- Lecturers can scan student QR codes and view their own scanning history, while students only see member screens.

Setup and deployment

- Run the SQL from supabase/schema.sql in the Supabase SQL Editor to create profiles, attendance, RLS policies, triggers, and avatar storage settings.
- Create lib/config.ts from lib/config.example.ts and add the Supabase URL and anon key.
- Create the first admin account from Supabase Dashboard Authentication → Users and include metadata like {"full_name": "Jane Doe", "role": "admin"}.
- Deploy the create-user Edge Function with the Supabase CLI so admin-created student and lecturer accounts work without exposing service-role credentials.
- Install dependencies with npm install and run npx expo start.

Database model

- profiles stores user identity, full name, role, avatar URL, and creation time.
- attendance records each scan event with the student user ID, date, recorder ID, and recorder role.
- The unique constraint is scoped to (user_id, date, recorded_by_role), so an admin and lecturer can both record the same student on the same day.
- RLS allows self-view, admin access, and lecturer access to student records needed for scanning and personal history.

Current implementation notes

- Profile pictures are admin-managed and stored in a public avatars bucket.
- The onboarding flow uses local image assets from assets/images/onboarding/.
- The app uses a single light theme defined in constants/theme.ts and avoids Tailwind or class-based styling.
- Attendance scans are unique per role and date to prevent duplicate entries while preserving separate admin/lecturer records.

Key files to know

- `app/_layout.tsx` — root navigator and role-based route guarding
- `app/(admin)/_layout.tsx` — admin tab layout
- `app/(member)/_layout.tsx` — member tab layout
- `hooks/use-auth.tsx` — shared authentication and profile resolution
- `services/attendance.ts` — attendance history, recording, and deduplication logic
- `services/admin.ts` — admin-side account creation through the Edge Function
- `supabase/schema.sql` — canonical database and RLS setup
- `app.json` — Expo app metadata and permissions

Quick start commands

```
npm install
npx expo start
npx supabase functions deploy create-user
```

This guide reflects the current repository state as of 2026-07-29.